Strengthening Data-driven Health and Life Sciences in the Netherlands: the role of TDCC-LSH

TDCC-LSH
Life Sciences & Health

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Thematic Digital Competence Centre for Life Sciences & Health

The Thematic Digital Competence Centre for Life Sciences and Health (TDCC-LSH) was established in September 2022 to advance digital practices within the life sciences and biomedical/health sciences in the Netherlands. TDCC-LSH is coordinated by Health-RI and one of three national TDCCs, alongside Natural & Engineering Sciences (NES) and Social Sciences & Humanities (SSH). TDCC-LSH aims to strengthen, harmonize and embed digital practices across the life sciences and biomedical/health sciences domains. The TDCC-LSH programme is based upon the TDCC-LSH Roadmap that has been formulated by the LSH research community.

The ambitions of TDCC-LSH are to
- Harmonise data stewardship and data access practices
- Enhance interoperability of digital solutions and resources
- Strengthen capacity and expertise in digital research

TDCC-LSH Strategy

- Targeted engagement of domain-specific data professionals across data driven health and life sciences fields
- Collaborative partnerships with major LSH stakeholders, initially most prominently the national Large Scale Research Infrastructures (incl Health-RI), BioSB and ELIXIR-NL
- Dual approach: Organisational and community level involvement
  - ✓ Top-down: Engage stakeholders at the institutional and organizational level to align strategic goals and resources
  - ✓ Bottom-up: Actively connect with domain-specific data stewards and expert communities to address grassroots needs and facilitate practical adoption
- Cross-disciplinary collaboration with TDCC-SSH and TDCC-NES, as well as with the Local Digital Competence Centers (LDCCs) at the RPOs
- Position the TDCC-LSH programme in the framework of the transition to Open Science

Network activities & workshops

- Data Steward Interest Group (DSIG) meetings
- National networking events & workshops
- Interview series "Spotlight on: Data Stewards"

https://tdcc.nl/category/spotlight-on/

Network & Stakeholders

Developing, strengthening and connecting networks and initiatives in data driven across life science & health domains

RPOs	*Research performing organisations (e.g. Universities, UMCs, research institutes, HBOs including their LDCCs.)*
RSOs	*Research supporting organisations (e.g. SURF, DANS, NLeSC, 4TU.ResearchData, etc.)*
National Infrastructures	*National domain-specific research infrastructure initiatives (e.g. health & life sciences LSRI-projects)*
International Infrastructures	*National node teams of European infrastructure initiatives (e.g. health & life sciences ESFRIs, EOSC, etc.)*
Data stewards	*Domain-specific data stewards and other professionals in data driven health and life sciences*
Policy Drivers	*Policy Drivers of digital technology, services and infrastructure*
TDCC-NES & TDCC-SSH	*Harmonisation with other TDCC domains*

TDCC-LSH projects 2025-2027

https://tdcc.nl/projects/tdcc-lsh-project-initiatives/

1. FAIR fellowship programme: Strengthening the Dutch LSH FAIR data stewardship landscape – installing a national cohort of FAIR LSH fellows – *Project Lead: Health-RI*
2. FAIR tool framework for bioinformatics services, tools and workflows in digital LSH research
   Project Lead: VU Amsterdam
3. LEARN-FAIR: LSH Educational Alignment for Research and Networking in FAIR Data Management
   Project Lead: Amsterdam UMC
4. Training the next generation of FAIR-aware and AI-savvy data scientists for the LSH domain
   Project Lead: Amsterdam UMC
5. Prospective harmonization of variables through Common Data Elements – towards a core cohort dataset for cohort studies in the Netherlands – *Project Lead: UMC Groningen*
6. FAIRify your metabolomics data: achieving convergence on standards for reuse-ready data and workflows – *Project Lead: Leiden University*

TDCC-LSH Accomplishments 2022-2024

1. Strong working relationships with all major Life Science & Health stakeholders; the field confirms the need for funding instruments that support digital infrastructure and practices
2. Six community-driven projects start(ed) in 2025; Project themes align with TDCC-LSH roadmap goals including FAIR principles, workflows, metadata, interoperability, capacity building, training, and data stewardship
3. Dutch Data Steward Interest Group (DSIG) strengthened, with an expanded scope and new domain-specific community meetings
4. Community engagement: a large number of TDCC-LSH (co-) organised events and outreach activities

TDCC-LSH Outlook

🔄 Transitioning from foundation to implementation

🤝 Deepening collaboration with stakeholders:
Solutions will be pursued both through funded TDCC-LSH projects and through targeted (network) activities in collaboration with domain-specific partners.

🧭 Further strategic positioning of TDCC-LSH in the Dutch data landscape, under guidance of the TDCC-LSH Programme Board

♻️ Exploring sustainability for the future:
In 2025–2026, TDCC-LSH will work with the broader LSH field to develop sustainable models for the continuation and long-term impact of the programme.

NWO · health RI enabling data driven health & life sciences · elixir NETHERLANDS · BioSB research school · TDCC-LSH Life Sciences & Health

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